

Operational resiliency through AI-assisted knowledge acceleration

Bridging the Gap: Using Generative AI and automated tooling to empower the next generation of Z talent, making the mainframe more consumable

IBM Resiliency Summit – New York August 2026

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From the big picture to one incident

01

Where application resiliency sits

Across the three pillars of this summit — and why it is the critical one

02

The approach

Before the incident, during it, and after — and what each phase must produce

03

IBM Bob Premium Package for Z

What it is, and which capability does the work in each phase

04

Scenario demo, and where you are today

One migration incident, run two ways — then a short self-assessment

Application resiliency runs through all three pillars

Hardware keeps the platform alive. Cyber keeps the data trustworthy. Neither tells you whether the business function is back.

Strategic foundations & hardware resilience

The big picture, and the physical infrastructure behind 24/7 operations



Cyber resiliency & recovery operations

Defending against active threats and restoring service rapidly



Intelligent operations & modernization

AI-driven management and future-proofing the ecosystem



Application resiliency

The layer all three depend on

A platform can be up, clean and modern while the business process is still down. Only the application layer tells you which one you are looking at.

PART ONE

Application resiliency

Where the outage clock is actually spent

The infrastructure recovers in minutes. The knowledge takes hours.

Minutes

to restart a subsystem

Infrastructure does its job

Hours

to establish what depends on it

No documented dependency path

Days

to be confident it will not repeat

Findings live in one engineer's notes

Those hours are an application intelligence problem — which is why this layer, not the platform, sets your real recovery time.

Resiliency is decided before the incident, not during it

One structure, three phases, three sets of artifacts that outlive the event.

BEFORE

Prepare & prevent

“What would break, and how would we know?”

Application dependency map

JCL risk inventory

Tier-1 recovery runbooks

DURING & UPTO RESOLUTION

Respond & contain

“What is in scope, and what happens if I touch it?”

Blast radius in minutes

Plain-language triage

Sequenced restart plan

AFTER

Learn & harden

“What stops this happening again?”

Contributing factor register

Optimization backlog

Modernization fact base

Build the map while nothing is on fire

Discovery moves out of the war room and into the planning window.

WHAT THE PHASE MUST PRODUCE

Application dependency map	Programs, jobs, tables and transactions — including the distributed applications reaching in through Db2
JCL risk inventory	Hardcoded subsystems, spool exposure and unbounded regions, with member and step references
Tier-1 recovery runbooks	Restart order and verification steps, drafted from source and validated by your SMEs

WHAT IT PREVENTS

Blast radius discovered under outage pressure
Defects that trigger at the first workload peak after cutover
Procedures written for the first time during the outage

Blast radius before workaround

The response has a fixed shape. Each step answers one question, in order.

01

Blast radius

Who fails, who survives, who needs restarting

>

02

Failing component explained

Purpose, data touched, why it abended, what to do

>

03

Abnormal workload traced

From the symptom to the line of code causing it

>

04

Change impact assessed

Before the workaround is applied, not after

The order is the discipline: scope the blast radius before you change anything.

The fix for the incident becomes the baseline

Contributing factor register

Every factor, the gap it exposed

Owner and target date

Nothing closed as “multi-factor”

Optimization backlog

The CPU and Db2 patterns the peak found

Ranked by incident impact

Quick wins separated out

Modernization fact base

Disposition per application

Complexity and knowledge risk

Evidence from code, not interviews

The baseline is stronger after every event. That is the difference between resilience and luck.

PART TWO

IBM Bob Premium Package for Z

The capability that makes the approach practical

Most AI coding tools see the file. Premium Package for Z sees the enterprise.

A system, not another assistant — it plans, executes, validates and governs multi-step work.

IBM Bob Premium Package for Z

Bob PPZ differentiates by grounding AI in enterprise Z metadata, domain context, client dictionaries, and runtime evidence

1

Enterprise scale understanding

Reduces blind spots across applications, dependencies and plans.

2

Mainframe modernization workflows

Connects analysis, transformation and validation.

3

Z application domain expertise

Keeps guidance platform-compliant and governance-aligned.

4

Enterprise data dictionaries

Turns cryptic variables into business meaning.

5

Runtime awareness

Targets active paths using execution evidence.

Outcome

Business-aware modernization decisions

Two modes: one plans the change, one makes it

Z Architect mode

Plan and analyse before anything is implemented

ASKED LIKE THIS

“What breaks if I change this field?”

Z Code mode

Explain, generate, refactor and transform code

ASKED LIKE THIS

“Add the column and update every dependent program”

The same assistant your distributed teams already understand — pointed at the platform they do not.

Which capability does the work in each phase

One analysis engine, pointed at three different questions.

BEFORE

Application analysis — dependency extraction across programs, jobs and data

JCL static analysis — migration-sensitive patterns

Documentation generation — runbook drafts from source

DURING & UPTO RESOLUTION

Code explanation — what this program does, in plain language

Change impact analysis — what moves if I apply this

On-demand dependency query — natural language, against the source

AFTER

Z code optimization advice — CPU and Db2 access findings

Business rule extraction — logic recovered from the code

Fact base generation — disposition and complexity per application

Where it stops: Bob reasons about what the code does — not about when your operations team allows things to run. Every artifact is validated before use.

PART THREE

Scenario demo

One migration incident, run two ways — and what separates the two runs

A managed-service migration, two weeks into hypercare

Cutover was clean. Nothing that failed had been changed — everything that failed depended on something that was.

The trigger

Infrastructure migration
Quarter-end workload peak
Priority 1 declared, seven sites affected

What failed

Distributed applications lost Db2 connectivity
IMS transactions abending
Spool filling, batch CPU spiking

The application

Plant-floor work order suite
IMS, CICS, Db2 and batch
Distributed consumers reaching in

Eight contributing factors. Five were already visible in code the client owned.

Hardcoded subsystem in JCL · spool management absent · unbounded sort region · unbounded Db2 cursor · row-by-row data access in a loop

Two are infrastructure faults. The application layer does not repair them — it changes when you know.

The same incident, run twice

Same team, same skills, same infrastructure. One had the baseline; one did not.

Without the baseline	72 hours to service · reopened once
With the baseline	1 h 35 m to service · no reopen

RESPONSE STEP	WITHOUT	WITH BOB PPZ
Blast radius established	2 h 15 m	4 min
Failing transaction explained	90 min waiting on one engineer	3 min
Factors still unidentified at closure	5 of 8	0 of 8
Steps blocked waiting on an SME	8	1

Both runs needed the same coupling-facility fix. In the fast run it was a planned change; in the slow run it was discovered at hour forty.

This is what makes the platform consumable to teams

STEPS BLOCKED WAITING ON AN SME

8 → 1

The expert still validates every artifact. They are no longer the queue.

Your SMEs review instead of derive

Two minutes checking a drafted answer, not ninety deriving it — and nobody waits

New joiners onboard against evidence

Documentation generated from the code, not folklore passed down verbally

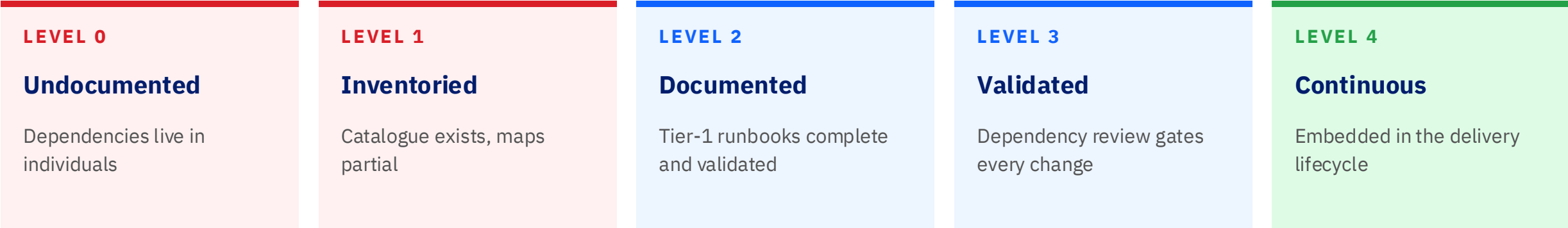
The knowledge outlives the individual

Retirement and turnover stop being resiliency events

The expertise is not replaced. It is made available at the speed operations actually runs at.

Where are you today?

Most estates sit in the first two — regardless of how good the infrastructure is.



THREE QUESTIONS THAT PLACE YOU HONESTLY

- 1 Could you map your three most critical applications in a week — today, without hiring?
- 2 Could a non-SME restart a Tier-1 application using only documentation?
- 3 Does a dependency review gate your next infrastructure change?

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Thank you

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